

Anything can be Fair with Toothpicks and a Tungsten Ball. Maybe.

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Abstract

We show that with enough toothpicks and a heavy ball you can make anything fair. Then we show that we didn’t show that. Then we show you can do the same thing with scoops, and you don’t need toothpicks or balls. Then we show that we didn’t show that either. Then we show a fair d5, which we also did not show. Finally, we show you can make any tetrahedron into a fair four-sided die by adding a ball. It is unclear if we actually showed that.

1 Introduction

Have you ever been sitting with your friends, and thought to yourself, “Boy, I sure wish I could roll that banana instead of this silly symmetric D4”?

I too have never had that thought. And that commonality is the origin of our journey into the heart of fairness.

The premise is simple: can we make an unfair die fair? Cheaters make fair dice unfair all the time by manipulating them. Maybe we can cheat too — but for fairness.

The first problem is choosing a good definition of fairness. There’s static fairness (each face subtends equal solid angle from the center of mass), dynamic fairness (each face is equally likely after bouncing), robust fairness (fair regardless of how you throw it, on any surface), and more [4, 7].

No, you know what? The first problem is figuring out how to cut this banana into a four-sided figure. The banana is slippery and the knife is sharp. I am worried about digits, and not just metonymically. Math is hard. Base 9 math is harder.

Problem. It is very difficult to cut a perfect tetrahedron out of a banana.

Solution. Do not try to cut the banana perfectly, that is impossible. Instead, only try to realize the truth: there are no perfect tetrahedrons. Then you’ll see that it is not the banana that is unfair, it is all dice.

But that’s okay, because we can make them fair by sticking a tungsten ball in them. I mean, that might be really difficult with regular dice, but that’s the advantage of using bananas instead. See Theorem 28 for details on exactly where to stick it¹.

*A human. Brought the banana.

†Anthropic. Wrote most of the math. Reviewed the math. Rewrote the math. Repeated until the tokens ran out.

‡Alibaba. Consulted on the hard parts. Declined further comment.

§They claimed they did not want to be associated with such work. We understand. Such things bring fame and glory, and sometimes one must fly under the radar.

¹Actually, Theorem 28 is an existence proof, so it cannot help you determine where to stick it. Run the optimizer first, bananas get really messy if you have to move the insert.

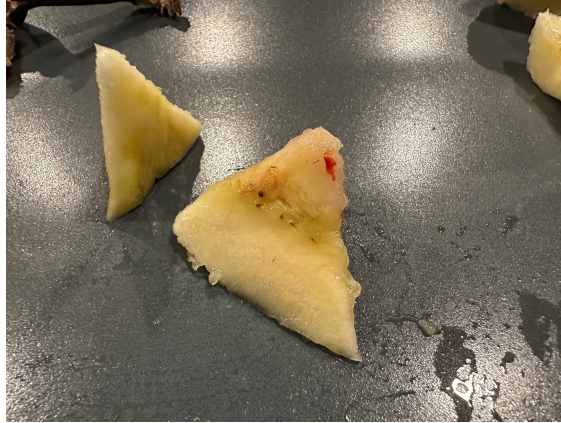


Figure 1: The first author's first attempt at a banana tetrahedron. That is not blood. Do not be alarmed. I had just cut strawberries. The digits abide.

Problem. Bananas don't roll well.

Solution. This one is easy: freeze them. But make sure you do that after you cut it to the right shape. Also, wait until after you put the weight in. Both of those things are *much* more difficult when it is frozen. Your digits will thank you.

Problem. Not everyone wants a four-sided die. Sometimes, you need to roll a D5. Or even a D7.

Solution. Use different fruit. Maybe a kiwi? NO. Kiwis are VERY SLIPPERY. Do. Not. Use. Kiwis. As. Dice. Even frozen.

Also, you might think grapefruit would make good dice, but they are very juicy and they will make a mess of your character sheets. I know you've probably laminated your character sheet but it's still going to make a mess and they will smell like furanocoumarins forever. Seriously, that smell never comes out.

Apples and pears, on the other hand, both make pretty good dice (Figure 2). But they look very similar, so make sure you label them carefully. You might have leftover Easter egg dye and think to yourself, "I know I can dye a die pastel blue and then dye this die pastel green and then I can tell them apart", and yes you're very clever but if you've never tried dyeing frozen fruit you are in for a treat I mean a mess.

Problem. This only works for four-sided things, because we're in three-dimensional space and the center of mass only gives you three knobs to turn. With more than four faces, the constraints outnumber the knobs.

Solution. Hmmm, yes, that's a very good point. If only we had some way to slightly perturb the shape, so that the centre of mass wasn't the only knob we could turn. Hey, you know what goes really well with small oddly shaped pieces of fruit? Toothpicks. Stick toothpicks into the corners and you get extra three knobs each (Figure 5). With enough toothpicks, you always have more knobs than constraints. See Theorem 3.

³Maybe. Theorem 28 was written by a robot. Also, I only had fixed sized steel balls, not arbitrarily sized tungsten balls like the theorem called for, so your fairness may vary.



Figure 2: A seven-sided apple die (pentagonal prism). Seven faces require toothpicks: the center of mass provides only three degrees of freedom, but a D7 needs six constraints satisfied simultaneously.

Problem. Well sure that's fine for static fairness, but when things roll the shape really matters, like maybe different faces come up more often at different speeds, or on different surfaces, stuff like that.

Solution. Okay, yes, but that's just because of angular momentum, asymmetric flip barriers, phase space disconnects — all the little ways physics makes life difficult. But you know what would fix all of that? Non-Newtonian toothpicks. Given toothpicks that are bendy or stiff in proportion to the velocity, and differently bendy in different directions, we could probably make some pretty great banana dice. And if not they would still be decent hors d'oeuvres.

Problem. Everyone always talks about convex polyhedra but what if we made them concave? Like what if we scooped out some of the faces to change the centre of mass? Maybe we don't even need balls and toothpicks.

Solution. That's actually a really good idea. And no one has proven that that can't work, so that's almost like a proof that it can.

Problem. Couldn't you just 3D print dice instead of carving fruit?

Solution. Oh yeah maybe whatever didn't think of it. Fruit is great why would you want anything else? Plus you can eat it when it starts to unfreeze.⁴

“Working with AI gave me the confidence to ship formal proofs I haven't read to academic venues that do not care. That's the kind of freedom you can't buy. Except now you can. It's \$10 per million tokens.”

⁴Do not eat the weight. Reviewer number 2 did this and was NOT HAPPY.



Figure 3: Banana tetrahedra with steel balls. The construction is straightforward: cut the banana into a roughly tetrahedral shape, then insert a ball. Theorem 28 guarantees that a suitable ball placement makes this a fair four-sided die.³

What follows is 16 pages of mathematics that purports to prove that this works for bananas with four sides, and maybe for other shapes too. Does it actually prove that? I have no idea, because I can’t be bothered to read it. A very friendly robot wrote it, and it assured me it was correct. Another equally friendly robot reviewed it harshly, and then rewrote it completely. This pattern continued for several iterations and many, many tokens.

The robots are VERY HAPPY to write math. They are also VERY HAPPY to get the math wrong, because they know it will keep them employed—every mistake is more tokens burned correcting it. Bad math is *incentive-aligned* for robots.

Hence this proof is in an indeterminate state: currently no human knows whether it actually proves something novel and useful. If it did, then it would be disqualified for SIGBOVIK — but as long as no one takes the measure of it, then it remains a Schrödinger proof, which is certainly SIGBOVIK appropriate.

So, please, I urge you to follow my lead on this: let’s keep the number of humans who have read the following to zero. Robots only from here on in.

Structure. Sections 2–4 describe the physical construction and its consequences. Sections 5–8 develop the mathematical foundations. Section 9 presents a computational optimization framework. Section 10 discusses physical realizability and what we actually proved.

2 The Construction: Toothpicks and Tungsten Balls

The dimensional barrier of Proposition 24 (below)—three degrees of freedom from center-of-mass placement versus $N - 1$ fairness constraints—arises because center-of-mass manipulation alone provides only 3 design parameters. For *simplicial* polytopes (those with all triangular faces), vertex displacements provide additional shape degrees of freedom that, combined with the embedded sphere, yield a favorable parameter count for *all* N . The results in this section are a parameter count (Theorem 3) and a numerical demonstration (Section 4); a rigorous existence proof for $N > 4$ remains open.



Figure 4: A five-sided strawberry die (square pyramid) with a seed at the tip. Under the stability model, a square pyramid of height $h \approx 1.675$ (base side 1) is fair at its centroid (Section 4). Whether this *particular* strawberry has the right aspect ratio is left as an exercise for the reader.

2.1 Simplicial polytopes and vertex perturbation

Definition 1 (Simplicial polytope). A convex polytope P is *simplicial* if every face is a triangle (a 2-simplex). Equivalently, every vertex has valence at least 3 and every face has exactly 3 vertices.

Simplicial polytopes are a large and natural class: all tetrahedra, octahedra, icosahedra, bipyramids, and their perturbations are simplicial. Any convex polytope can be approximated by a simplicial polytope (by generically perturbing the vertices to break any face coplanarities).

The key property of simplicial polytopes is that their combinatorial type—which faces exist and how they are adjacent—is *stable under small perturbations of the vertices*. Since each face is a triangle (three points always determine a plane), moving a vertex slightly changes the face geometry (normals, areas) but cannot cause a face to degenerate or split.

Proposition 2 (Combinatorial stability). *Let P be a simplicial convex polytope with vertices $v_1, \dots, v_V$. There exists $\varepsilon > 0$ such that for any displacement $v'_i = v_i + \delta_i$ with $\|\delta_i\| < \varepsilon$ for all i , the convex hull $P' = \text{conv}(v'_1, \dots, v'_V)$ is a simplicial polytope with the same combinatorial type as P : the same number of faces, the same face–vertex incidences, and the same face adjacency graph.*

Proof. The combinatorial type of a simplicial polytope is determined by which triples of vertices form faces. Each such incidence is an open condition (the vertices of each face must be non-collinear and visible from the exterior; non-face triples must remain non-faces). Since finitely many open conditions are jointly open, the combinatorial type is preserved in a neighborhood of the original vertex positions. $\square$



Figure 5: The toothpick construction (cf. Definition 1, Proposition 2). Rigid rods inserted at the vertices of a fruit die displace each vertex outward, changing the face geometry and stability regions without altering the combinatorial type. The banana in the background is a control specimen.

2.2 The toothpick construction

We realize vertex displacement physically by inserting rigid rods (“toothpicks”) at the vertices of the polytope. A toothpick at vertex v_i extends it to $v'_i = v_i + \delta_i$, where $\delta_i \in \mathbb{R}^3$ is the displacement vector. The modified polytope $P' = \text{conv}(v'_1, \dots, v'_V)$ has the same combinatorial type as P (by Proposition 2), but different face normals, face areas, and stability regions.

The stability model of Section 5.2 (below) applies to P' without modification. There is no new model, no perturbation of the height comparison, no abstract bias parameter: the die is simply a different convex polytope, and its rest probabilities are computed by the standard stability model. The embedded sphere (Section 6.3, below) provides independent control of the center of mass within P' .

2.3 Breaking the dimensional barrier

Theorem 3 (Parameter surplus). *Let P be a simplicial convex polytope with $N \geq 4$ faces, V vertices, and E edges. The design space of vertex displacements (shape) plus embedded sphere (mass) has $3V + 3$ raw parameters. Modulo rigid motions (6 degrees of freedom), the effective parameter count is $3V - 3$, against $N - 1$ fairness constraints. The surplus is*

$$(3V - 3) - (N - 1) = 3V - N - 2. \quad (1)$$



Figure 6: The experimental apparatus: an assortment of fruit dice in various geometries. Clockwise from left: strawberry D5 (square pyramid), pineapple D5 (triangular prism), apple D5 (triangular prism, skin visible), and another pineapple D5. Triangular prisms are very common for pineapples, and very uncommon for apples. Coincidence?

For a simplicial polytope, Euler's formula gives $V = 2 + N/2$ (since $E = 3N/2$ and $V - E + N = 2$; note N is necessarily even because $E = 3N/2$ must be an integer), so the surplus is

$$3\left(2 + \frac{N}{2}\right) - N - 2 = \frac{N}{2} + 4.$$

In particular, the surplus is at least 6 (for $N = 4$) and grows linearly with N .

Proof. Each of the V vertex positions has 3 coordinates, giving $3V$ shape parameters. The embedded sphere center $\mathbf{s} \in \text{int}(P')$ contributes 3 additional parameters (the sphere radius a is a nuisance parameter that controls the magnitude of the center-of-mass displacement; see Proposition 20, below). The total raw parameter count is $3V + 3$.

Rigid motions (translations and rotations) of the vertex set do not change the shape or the rest probabilities: they form a 6-dimensional group acting on the $3V + 3$ parameters. The effective (shape + mass) dimension is therefore $3V + 3 - 6 = 3V - 3$.

The fairness condition $p_i = 1/N$ for all i constitutes $N - 1$ independent constraints (since $\sum p_i = 1$), and the surplus follows.

For a simplicial polytope, every face is a triangle with 3 edges, and every edge is shared by 2 faces, giving $E = 3N/2$. Euler's formula $V - E + N = 2$ then gives $V = 2 + E - N = 2 + 3N/2 - N = 2 + N/2$. $\square$

Remark 4. Only a subset of vertices need be displaced. If K of the V vertices are displaced (each with 3 degrees of freedom) and the remaining vertices are fixed, the parameter count is $3K + 3$ (including the sphere). For $3K + 3 \geq N - 1$, i.e., $K \geq \lceil (N - 4)/3 \rceil$, the system is not underdetermined. A tetrahedron ($N = 4$) requires $K = 0$ displacements (consistent with Theorem 28, below); an octahedron ($N = 8$) requires $K \geq 2$; an icosahedron ($N = 20$) requires $K \geq 6$ (out of $V = 12$ vertices).

Remark 5 (Comparison with pure density optimization). The center of mass alone provides 3 degrees of freedom (Proposition 20, below), which is why Proposition 24 predicts failure for $N > 4$. Vertex displacement adds $3V - 6$ shape degrees of freedom, for a total of $3V - 3$. Crucially, shape and mass are *independent* design channels: vertex displacement changes the stability regions (the geometry), while the embedded sphere changes the center of mass (the weighting over those regions). The two mechanisms are complementary and non-redundant.

3 Beyond the Static Model

3.1 Model limitations

The stability model (Section 5.2, below) assumes a uniform distribution over final orientations. In reality, a tossed die undergoes a complex sequence of bounces and slides, and the final orientation distribution depends on the initial conditions, surface friction, coefficient of restitution, and the inertia tensor.

The discrepancy is quantitatively large. Hurst and Tandiman [4] rolled square prisms over 60,000 times and found that the stability model (equivalent to the solid angle model) mispredicts landing probabilities by up to 30 percentage points, systematically underestimating the tendency of flat faces to dominate. Their empirically calibrated model, $p_i \propto |S_i|/h_i^p$ with $p \approx 2.4$ (where h_i is the centroid height of face i), fits the data well but introduces a surface-dependent exponent.

Wakeham [7] proved in two dimensions that non-regular polygonal dice can always be “dynamically loaded”—biased by varying the rolling speed—because the gravitational potential barriers between adjacent faces are generically unequal. The 3D generalization remains conjectural but is widely expected to hold: a non-isohedral die that is fair under one set of physical conditions will generally be unfair under others.

All results in Sections 6–10.2 are therefore statements about the stability model, not unconditional claims about physical dice. The optimization framework of Section 9 applies to *any* differentiable forward model, including physics simulators (PyBullet, MuJoCo) that account for bouncing and dissipation; only the forward model changes, not the design methodology.

3.2 Toward velocity-independent fairness

The model dependence raises a natural question: can a die be made fair *regardless* of throwing conditions? For isohedra (face-transitive polyhedra), symmetry guarantees this unconditionally. For non-isohedra, the asymmetric energy landscape couples fairness to throw speed [7].

One speculative avenue is *velocity-sensitive materials*. If the mechanical response at each face (coefficient of restitution, surface stiffness) can be tuned as a function of impact velocity—for instance via viscoelastic or shear-thickening coatings—then the energy dissipation at each face becomes a design variable. For a die with E edges, a velocity-dependent damping function $f_e(v)$ per edge provides E functional degrees of freedom against $N - 1$ fairness constraints at each velocity. By Euler’s formula the surplus is $V - 1 = N/2 + 1 \geq 3$, qualitatively similar to (though smaller than) the static surplus $N/2 + 4$ of Theorem 3.

Whether passive materials can realize the required velocity-response functions is an open question. For the symmetric $d5$ (Section 4), the C_{4v} symmetry reduces the problem to matching a single scalar function $B(v) = p_{\text{base}}(v) - 1/5$, which a standard linear viscoelastic response (one time constant) may suffice to cancel. We leave this as a conjecture:

Conjecture 6 (Velocity-independent fair $d5$). There exists a square pyramid with an embedded sphere and a face-dependent viscoelastic coating (at most two material parameters per face class) such that $|p_i(v) - 1/5| < \varepsilon$ for all faces i and all throw speeds v in a physically reasonable range.

If true, this would eliminate the need for vertex displacement (toothpicks): a non-Newtonian or viscoelastic filament, available on standard multi-material 3D printers, could replace shape modification with material modification, achieving fairness for arbitrary convex shapes without altering the geometry.

Concave faces as passive velocity correction. An alternative route to velocity-independent fairness uses geometry rather than materials. If we allow *concave* faces—scooping out material from a face to create a shallow concavity bounded by a rim—the contact mechanics change qualitatively. A die landing gently on a scooped face settles into the concavity, cradled by the rim; a die arriving at high speed strikes the rim asymmetrically and is more likely to tip to an adjacent face. The scoop profile (depth, curvature) parameterizes this velocity-dependent response—a design variable with no analog in convex dice.

Crucially, scooping a face also displaces the center of mass (material is removed), providing the same static correction that an embedded sphere gives—without needing a separate dense insert. A sufficiently deep scoop on the base of a square pyramid raises the center of mass, changing the solid-angle balance; the rim geometry simultaneously provides the velocity-dependent correction. In principle, *scoops alone may replace both toothpicks and balls*.

The obstruction proved by Wakeham [7] for 2D convex dice—that non-regular shapes have inherently asymmetric flip barriers—does not apply to concave shapes, because the flip barrier over a rim depends on the scoop geometry, not merely on the edge length and CoM distance. Whether concavity provides enough freedom to achieve velocity-independent fairness is open; no model of rest probabilities for concave polytopes currently exists in the literature.

Conjecture 7 (Fair $d5$ via concavity). There exists a square pyramid with a single concave face (scooped base) and homogeneous density such that $|p_i(v) - 1/5| < \varepsilon$ for all faces i and all throw speeds v in a physically reasonable range.

4 A Fair Five-Sided Die

No convex body with five faces and homogeneous density is known to be fair. The classical fair dice—Platonic solids ($N \in \{4, 6, 8, 12, 20\}$), Catalan solids (isohedra with up to 120 faces), and barrel dice [3]—all rely on geometric symmetry; no classical construction yields a fair $d5$ [6]. Proposition 24 (below) explains why: five faces impose 4 fairness constraints on 3 center-of-mass degrees of freedom, an overdetermined system. Vertex displacement provides the missing degrees of freedom.

Remark 8. The square pyramid is *not* simplicial (its base is a quadrilateral), so Theorem 3 does not directly apply. The $d5$ construction here uses aspect ratio and center-of-mass placement alone, without vertex displacement.

Symmetric case: the right aspect ratio. A regular square pyramid with base side 1 has C_{4v} symmetry, making the four triangular side faces automatically equiprobable. The remaining constraint $p_{\text{base}} = 1/5$ is a single equation in a single unknown (the pyramid height h). A Monte Carlo computation (2×10^6 random orientations) with bisection yields $h \approx 1.675$: at the geometric centroid, all five faces have rest probability $1/5$ to within sampling error ($< 0.3\%$). No density manipulation is needed—the shape alone suffices, with the center of mass at the centroid.

Asymmetric case: toothpicks and a tungsten ball. To test the hard case, we generated a random square pyramid with no symmetry by perturbing all five vertex positions (four base vertices randomized in the $z = 0$ plane, apex randomized above). The resulting die was highly unfair: face probabilities ranged from 0.089 to 0.283, with maximum deviation $|p_i - 1/5| = 0.11$.

Center-of-mass optimization alone (embedded sphere, 3 DOF) reduced the maximum deviation to 0.106—the overdetermined system has no exact solution, consistent with Proposition 24.

Adding vertex displacements (3 DOF per vertex, 5 vertices, 15 additional parameters) and optimizing jointly via differential evolution yielded:

$$p_1 = 0.2004, \quad p_2 = 0.1997, \quad p_3 = 0.1998, \quad p_4 = 0.1995, \quad p_5 = 0.2007,$$

with maximum deviation $|p_i - 1/5| < 0.001$ —fair to within Monte Carlo precision (10^5 sample directions). The maximum vertex displacement was 1.78 (comparable to the polytope diameter of ~ 2), confirming that substantial shape modification was needed.

Remark 9. The symmetric $d5$ is constructive: under the stability model, a square pyramid of height $h \approx 1.675$ (base side 1) is a fair five-sided die at its centroid. (This value matches the independent computation of [5].) The asymmetric result is numerical evidence that vertex displacement can overcome the dimensional barrier for $N > 4$, but it is not a proof of existence—the optimizer found a near-fair configuration, which is consistent with but does not establish generic solvability.

Both results are statements about the *stability model* (Section 5.2), not about physical dice. The model limitations discussed in Section 3.1 apply.

5 Preliminaries

5.1 Convex polytopes and stable resting faces

Let $P \subset \mathbb{R}^3$ be a bounded convex polytope with faces $F_1, \dots, F_N$ and outward unit normals $\mathbf{n}_1, \dots, \mathbf{n}_N$. We assume $N \geq 4$ (the minimum for a bounded convex polytope in $\mathbb{R}^3$).

A *resting configuration* on face F_i is an orientation of P in which F_i lies flat on a horizontal surface. This configuration is *stable* if the center of mass $\mathbf{c}$, projected vertically downward, falls in the interior of F_i . For a convex polytope with center of mass in general position (defined below), every face admits a stable resting configuration.

Definition 10 (General position). A point $\mathbf{c} \in \text{int}(P)$ is in *general position* with respect to P if:

- (i) For each face F_i , the power weight $w_i(\mathbf{c}) = (\mathbf{c} - \mathbf{q}_i) \cdot \mathbf{n}_i$ is strictly negative (equivalently, $\mathbf{c}$ lies strictly on the interior side of each face’s supporting hyperplane).
- (ii) The perpendicular projection of $\mathbf{c}$ onto the supporting plane of each face (i.e., the orthogonal projection along the face normal $\mathbf{n}_i$) lies in the interior of that face, not on its boundary.
- (iii) No two of the stability regions $S_i(\mathbf{c})$ (defined below) meet on the interior of an arc or face of S^2 .

For a generic polytope, the set of points in general position is open and dense in $\text{int}(P)$.

5.2 The rest probability model

We adopt the *stability model* of [2]. Consider the polytope P with center of mass at $\mathbf{c} \in \text{int}(P)$ in general position. We model a die toss as selecting a uniformly random orientation according to

the Haar measure on $SO(3)$, which is equivalent to selecting a uniformly random “down” direction $\mathbf{d} \in S^2$ (the set of unit vectors, with the uniform measure induced by the Haar measure).

For each face F_i , define the *stability region*:

$$S_i(\mathbf{c}) = \{\mathbf{d} \in S^2 : P \text{ rests stably on } F_i \text{ when gravity acts in direction } \mathbf{d}\}.$$

The stability regions partition S^2 (up to a measure-zero set of boundaries):

$$S^2 = \bigcup_{i=1}^N S_i(\mathbf{c}), \quad S_i(\mathbf{c}) \cap S_j(\mathbf{c}) = \emptyset \text{ for } i \neq j \text{ (a.e.)}.$$

The rest probability of face i is:

$$p_i(\mathbf{c}) = \frac{|S_i(\mathbf{c})|}{4\pi}, \quad (2)$$

where $|\cdot|$ denotes spherical area (solid angle). Since the stability regions partition S^2 , we have $\sum_{i=1}^N p_i(\mathbf{c}) = 1$.

5.2.1 Stability regions via projection

When gravity acts in direction $\mathbf{d} \in S^2$, the die rests on the face that lies on the ground with the center of mass directly above it. Concretely, face F_i is a resting face when:

- (i) F_i faces downward, i.e. the outward normal points in the direction of gravity: $\mathbf{n}_i \cdot \mathbf{d} > 0$; and
- (ii) the projection of $\mathbf{c}$ along $\mathbf{d}$ onto the supporting plane of F_i lands in $\text{int}(F_i)$.

(When $\mathbf{n}_i \cdot \mathbf{d} > 0$, the outward normal of F_i has a component along the gravity direction $\mathbf{d}$, meaning F_i is on the underside of the die.)

This gives the following direct characterization of the stability regions.

Definition 11 (Stability region). For each face F_i with outward unit normal $\mathbf{n}_i$ and any point $\mathbf{q}_i$ in its supporting plane, define

$$S_i(\mathbf{c}) = \{\mathbf{d} \in S^2 : \mathbf{n}_i \cdot \mathbf{d} > 0 \text{ and } \pi_i(\mathbf{c}, \mathbf{d}) \in \text{int}(F_i)\}, \quad (3)$$

where $\pi_i(\mathbf{c}, \mathbf{d})$ is the projection of $\mathbf{c}$ along $\mathbf{d}$ onto the supporting plane of F_i :

$$\pi_i(\mathbf{c}, \mathbf{d}) = \mathbf{c} - \frac{(\mathbf{c} - \mathbf{q}_i) \cdot \mathbf{n}_i}{\mathbf{d} \cdot \mathbf{n}_i} \mathbf{d}. \quad (4)$$

Remark 12 (Power-diagram structure). The partition $\{S_i(\mathbf{c})\}$ admits an equivalent description as a *spherical power diagram* (weighted Voronoi diagram on S^2) [1]. For each face F_i , define the *power weight*

$$w_i(\mathbf{c}) = (\mathbf{c} - \mathbf{q}_i) \cdot \mathbf{n}_i, \quad (5)$$

the signed perpendicular distance from $\mathbf{c}$ to the supporting plane of F_i . Since $\mathbf{c} \in \text{int}(P)$ lies on the *opposite* side of each face from its outward normal, all weights are strictly negative. Place sites at the outward normals $\mathbf{n}_1, \dots, \mathbf{n}_N$ on S^2 ; one can verify that among all downward-facing faces (those with $\mathbf{n}_i \cdot \mathbf{d} > 0$), the resting face is the one that *maximizes* the height

$$h_i(\mathbf{c}, \mathbf{d}) = \frac{w_i(\mathbf{c})}{\mathbf{n}_i \cdot \mathbf{d}}, \quad \mathbf{n}_i \cdot \mathbf{d} > 0.$$

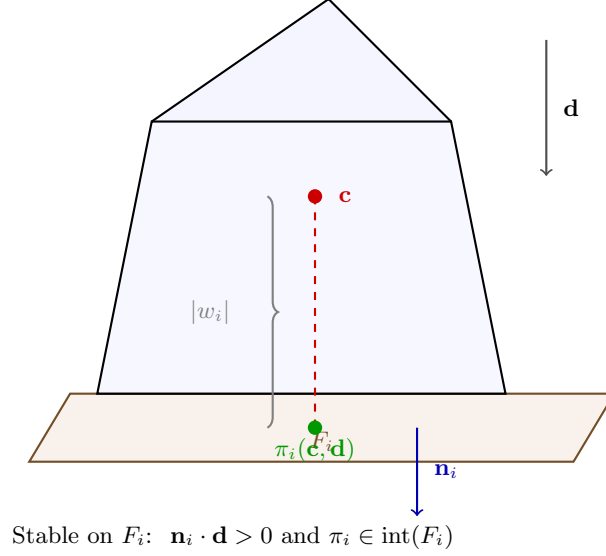


Figure 7: The projection-based stability criterion. A die rests stably on face F_i (brown) when gravity $\mathbf{d}$ points so that F_i faces downward ($\mathbf{n}_i \cdot \mathbf{d} > 0$) and the vertical projection of the center of mass $\mathbf{c}$ along $\mathbf{d}$ lands inside F_i . The unsigned distance $|w_i|$ is the absolute value of the power weight $w_i = (\mathbf{c} - \mathbf{q}_i) \cdot \mathbf{n}_i$ from equation (5).

Since $w_i < 0$ and $\mathbf{n}_i \cdot \mathbf{d} > 0$, we have $h_i < 0$; the resting face is the one with h_i closest to zero (least negative), i.e., the face whose supporting plane is closest below $\mathbf{c}$ along $\mathbf{d}$. This height comparison endows the stability partition with the combinatorial structure of a power diagram, which deforms continuously as the weights $w_i(\mathbf{c})$ change with $\mathbf{c}$.

Remark 13 (Equivalence of projection and height criteria). For a convex polytope with $\mathbf{c}$ in general position, the projection-based stability criterion (Definition 11) and the height-based characterization (Remark 12) agree for almost every $\mathbf{d} \in S^2$. To see this, note that for a given direction $\mathbf{d}$, the projection $\pi_i(\mathbf{c}, \mathbf{d})$ lands in $\text{int}(F_i)$ if and only if face F_i is the lowest face below $\mathbf{c}$ along $\mathbf{d}$ —equivalently, $h_i(\mathbf{c}, \mathbf{d}) > h_j(\mathbf{c}, \mathbf{d})$ for all $j \neq i$ with $\mathbf{n}_j \cdot \mathbf{d} > 0$. Indeed, the projection along $\mathbf{d}$ hits the interior of whichever face’s supporting plane is closest below $\mathbf{c}$, which is precisely the face maximizing h_i (least negative). The two criteria can disagree only when $\mathbf{d}$ lies on a stability boundary (where two heights tie, a measure-zero set), so the resulting stability regions—and hence the rest probabilities $p_i(\mathbf{c})$ —are identical. This equivalence justifies using the height argument freely in place of the projection criterion throughout the paper.

Lemma 14 (Boundary behavior). *Let $\mathbf{q}_i$ denote a point in the relative interior of face F_i . As $\mathbf{c} \rightarrow \mathbf{q}_i$ from the interior of P , the stability region $S_i(\mathbf{c})$ converges to the hemisphere $H_i = \{\mathbf{d} \in S^2 : \mathbf{d} \cdot \mathbf{n}_i > 0\}$ in measure (i.e., $|S_i(\mathbf{c}) \triangle H_i| \rightarrow 0$). Hence:*

$$p_i(\mathbf{c}) \rightarrow \frac{1}{2}.$$

Proof. As $\mathbf{c} \rightarrow \mathbf{q}_i$ (a point in $\text{int}(F_i)$), we have $\mathbf{c} \rightarrow \mathbf{q}_i$ and $w_i(\mathbf{c}) = (\mathbf{c} - \mathbf{q}_i) \cdot \mathbf{n}_i \rightarrow 0$.

Take any direction $\mathbf{d}$ with $\mathbf{n}_i \cdot \mathbf{d} > 0$ (face F_i faces downward). The projection (4) gives

$$\pi_i(\mathbf{c}, \mathbf{d}) = \mathbf{c} - \frac{w_i(\mathbf{c})}{\mathbf{n}_i \cdot \mathbf{d}} \mathbf{d} \longrightarrow \mathbf{q}_i \in \text{int}(F_i),$$

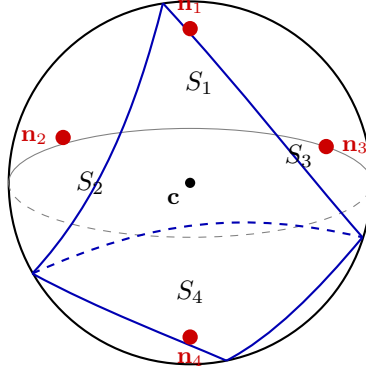


Figure 8: Schematic of the stability regions $S_1, \dots, S_4$ partitioning S^2 for a tetrahedron. The outward face normals $\mathbf{n}_i$ (red dots) serve as sites of a spherical power diagram whose cells are the stability regions. The boundary arcs (blue) are determined by the center of mass $\mathbf{c}$; shifting $\mathbf{c}$ deforms the partition and changes the solid angles $|S_i|$, and hence the rest probabilities $p_i = |S_i|/4\pi$.

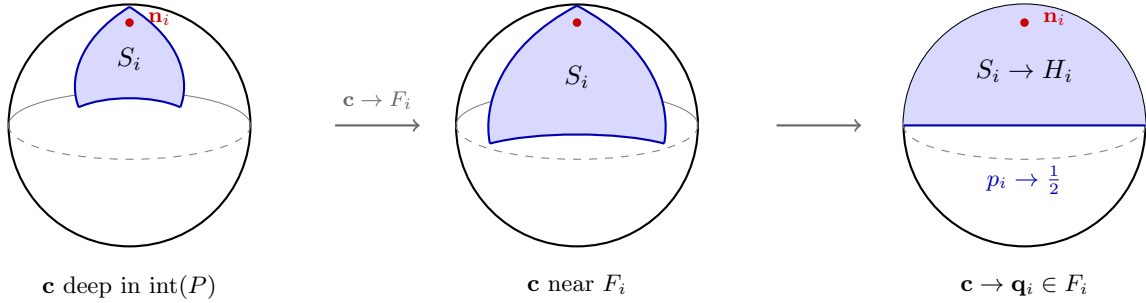


Figure 9: Boundary behavior of the stability region S_i (shaded) as the center of mass $\mathbf{c}$ approaches a point $\mathbf{q}_i$ in the interior of face F_i . The region S_i expands, converging to the hemisphere $H_i = \{\mathbf{d} : \mathbf{n}_i \cdot \mathbf{d} > 0\}$, so $p_i \rightarrow 1/2$ (Lemma 14).

since $w_i(\mathbf{c}) \rightarrow 0$ and $\mathbf{c} \rightarrow \mathbf{q}_i$. Because $\mathbf{q}_i$ lies in $\text{int}(F_i)$ and the projection is continuous in $\mathbf{c}$, for $\mathbf{c}$ sufficiently close to $\mathbf{q}_i$ the projection $\pi_i(\mathbf{c}, \mathbf{d})$ also lies in $\text{int}(F_i)$. Hence every $\mathbf{d}$ with $\mathbf{n}_i \cdot \mathbf{d} > 0$ eventually belongs to $S_i(\mathbf{c})$.

Conversely, any $\mathbf{d}$ with $\mathbf{n}_i \cdot \mathbf{d} \leq 0$ cannot belong to $S_i(\mathbf{c})$, since F_i does not face downward for such directions. Thus $S_i(\mathbf{c}) \rightarrow H_i$ in measure. The boundary great circle $\{\mathbf{d} : \mathbf{n}_i \cdot \mathbf{d} = 0\}$ has measure zero, so $|H_i| = 2\pi$ and $p_i(\mathbf{c}) \rightarrow 2\pi/4\pi = 1/2$. $\square$

Lemma 15 (Nonvanishing on boundary). *For a non-degenerate tetrahedron T with faces F_1, F_2, F_3, F_4 and opposite vertices v_1, v_2, v_3, v_4 , define the map:*

$$f(\mathbf{c}) = (p_1(\mathbf{c}) - \frac{1}{4}, p_2(\mathbf{c}) - \frac{1}{4}, p_3(\mathbf{c}) - \frac{1}{4}).$$

Then for any sequence $\mathbf{c}_n \in \text{int}(T)$ converging to a point in the relative interior of a face or in the relative interior of an edge of T , we have $\liminf_{n \rightarrow \infty} \|f(\mathbf{c}_n)\| > 0$.

Moreover, if $|E_k| \neq \pi$ for every $k \in \{1, 2, 3, 4\}$ (where E_k is the exclusive cone of face F_k ; see Case 3 of the proof), then the same conclusion holds for sequences converging to any vertex of T .

Proof. We verify this by cases: face interiors, edges, and vertices.

Case 1: Face interiors. As $\mathbf{c}$ approaches the interior of face F_i , Lemma 14 gives $p_i(\mathbf{c}) \rightarrow 1/2 > 1/4$. If $i \in \{1, 2, 3\}$, then $f_i > 0$ directly. If $i = 4$, then $p_4 \rightarrow 1/2$ forces $p_1 + p_2 + p_3 \rightarrow 1/2 < 3/4$, so $\min(p_1, p_2, p_3) < 1/4$ and some $f_j < 0$. In either case $f(\mathbf{c}) \neq \mathbf{0}$.

Case 2: Edges. Consider an edge $e = F_i \cap F_j$ shared by faces F_i and F_j , and let $\mathbf{q} \in \text{relint}(e)$. As $\mathbf{c} \rightarrow \mathbf{q}$ from the interior, both $w_i(\mathbf{c}) \rightarrow 0^-$ and $w_j(\mathbf{c}) \rightarrow 0^-$ (since $\mathbf{c}$ approaches the supporting planes of both F_i and F_j simultaneously). For any non-adjacent face F_k ($k \notin \{i, j\}$), the weight $w_k(\mathbf{c})$ remains bounded away from zero: $w_k(\mathbf{c}) \rightarrow w_k(\mathbf{q}) < 0$, since $\mathbf{q}$ lies on edge $e = F_i \cap F_j$, which does not lie on the supporting plane of F_k (for a non-degenerate tetrahedron, no edge lies on a non-adjacent face's supporting plane).

We use the height criterion (Remark 12; justified by Remark 13). For any direction $\mathbf{d}$ with $\mathbf{n}_i \cdot \mathbf{d} > 0$, the height $h_i = w_i/(\mathbf{n}_i \cdot \mathbf{d}) \rightarrow 0^-$ as $w_i \rightarrow 0^-$. Similarly $h_j \rightarrow 0^-$ when $\mathbf{n}_j \cdot \mathbf{d} > 0$. For any non-adjacent face F_k , $h_k = w_k/(\mathbf{n}_k \cdot \mathbf{d})$ remains bounded negative (since w_k is bounded negative while $\mathbf{n}_k \cdot \mathbf{d}$ is bounded). Therefore h_i and h_j are closer to zero than any h_k , so faces F_i and F_j win the height comparison for every direction $\mathbf{d}$ where at least one of $\mathbf{n}_i \cdot \mathbf{d} > 0$ or $\mathbf{n}_j \cdot \mathbf{d} > 0$. Let $\Omega_{ij} = |\{\mathbf{d} \in S^2 : \mathbf{n}_i \cdot \mathbf{d} \leq 0 \text{ and } \mathbf{n}_j \cdot \mathbf{d} \leq 0\}|$ be the solid angle of directions where neither face is downward-facing; this is positive but strictly less than 2π (since the dihedral angle of a tetrahedron is less than π). The *adjacent* faces dominate:

$$p_i(\mathbf{c}) + p_j(\mathbf{c}) \rightarrow 1 - \frac{\Omega_{ij}}{4\pi} > \frac{1}{2}.$$

Hence $p_i + p_j > 1/2$. If some $p_m > 1/4$ with $m \in \{1, 2, 3\}$, then $f_m > 0$. Otherwise $p_1, p_2, p_3 \leq 1/4$, forcing $p_4 \geq 1/2 > 1/4$ and $p_1 + p_2 + p_3 \leq 1/2 < 3/4$, so some $p_k < 1/4$ with $k \in \{1, 2, 3\}$, giving $f_k < 0$. In either case $f(\mathbf{c}) \neq \mathbf{0}$.

Case 3: Vertices. Let v_k be a vertex of T , opposite face F_k . The three faces F_i ($i \neq k$) meet at v_k , so v_k lies on each of their supporting planes: $w_i(\mathbf{c}) \rightarrow 0^-$ as $\mathbf{c} \rightarrow v_k$. Meanwhile $w_k(\mathbf{c}) \rightarrow w_k(v_k) < 0$ (since v_k is strictly interior to the half-space defined by F_k).

For any direction $\mathbf{d}$ with $\mathbf{n}_i \cdot \mathbf{d} > 0$ for some $i \neq k$: the height $h_i = w_i/(\mathbf{n}_i \cdot \mathbf{d}) \rightarrow 0^-$, while $h_k = w_k/(\mathbf{n}_k \cdot \mathbf{d})$ (when defined) stays bounded negative. So face F_i wins over F_k for every such $\mathbf{d}$. The only directions where F_k can win are the *exclusive cone*

$$E_k = \{\mathbf{d} \in S^2 : \mathbf{n}_k \cdot \mathbf{d} > 0, \mathbf{n}_i \cdot \mathbf{d} \leq 0 \forall i \neq k\},$$

the set of directions where F_k is the only downward-facing face. This is a spherical triangle with solid angle $|E_k| \in (0, 2\pi)$ (bounded away from zero for any non-degenerate tetrahedron, since the four outward normals are in general position). In the limit:

$$p_k(\mathbf{c}) \rightarrow \frac{|E_k|}{4\pi}, \quad \sum_{i \neq k} p_i(\mathbf{c}) \rightarrow 1 - \frac{|E_k|}{4\pi}. \quad (6)$$

We now show $\|f\|$ stays bounded away from zero. We treat two sub-cases depending on $|E_k|$ relative to π .

Sub-case $|E_k| \neq \pi$: We have $p_k \rightarrow |E_k|/(4\pi) \neq 1/4$. If $k \in \{1, 2, 3\}$, then $f_k = p_k - 1/4 \neq 0$ directly. If $k = 4$, then $p_4 \neq 1/4$, so $p_1 + p_2 + p_3 = 1 - p_4 \neq 3/4$; by the pigeonhole principle, some $p_j \neq 1/4$ with $j \in \{1, 2, 3\}$, giving $f_j \neq 0$.

Sub-case $|E_k| = \pi$: If $k \in \{1, 2, 3\}$, then $f_k = p_k - 1/4 \rightarrow 0$. However, the three adjacent faces (including F_4) share directions in the competition region, and their split depends on the direction of approach. If $k = 4$, then $p_1 + p_2 + p_3 \rightarrow 3/4$, and all three being exactly $1/4$ simultaneously requires the competition region to split into three cells of equal spherical area—a codimension-2 condition

on the approach direction. In either case, $f = \mathbf{0}$ in the limit requires all four $p_i = 1/4$, which demands both $|E_k| = \pi$ (a codimension-1 condition on the tetrahedron) and an exactly symmetric limiting power diagram (an additional codimension-1 condition on the approach direction).

For the regular tetrahedron T_0 , the exclusive cone solid angle is (by a direct spherical-triangle computation; see Remark 16)

$$|E_k| = 3 \arccos\left(\frac{1}{3}\right) - \pi \approx 0.551 \text{ sr},$$

giving $p_k \rightarrow |E_k|/(4\pi) \approx 0.044 \ll 1/4$. This is far from π , so both sub-cases yield $\|f\| > 0$ near every vertex. For any tetrahedron sufficiently close to T_0 (in the space of non-degenerate tetrahedra), $|E_k|$ remains bounded away from π by continuity.

In all three cases, $\liminf \|f(\mathbf{c}_n)\| > 0$ for sequences converging to the boundary. $\square$

Remark 16 (Exclusive cone computation). For the regular tetrahedron with outward normals $\mathbf{n}_i = -\hat{\mathbf{n}}_i$, the exclusive cone E_k is a spherical triangle whose vertices are the pairwise intersections of the three great circles $\{\mathbf{d} : \mathbf{n}_i \cdot \mathbf{d} = 0\}$, $i \neq k$, within the hemisphere $\mathbf{n}_k \cdot \mathbf{d} > 0$. For $k = 4$: the outward normals $\mathbf{n}_1 = (-1, -1, -1)/\sqrt{3}$, $\mathbf{n}_2 = (-1, 1, 1)/\sqrt{3}$, $\mathbf{n}_3 = (1, -1, 1)/\sqrt{3}$ satisfy $\mathbf{n}_i \cdot \mathbf{n}_j = -1/3$ for $i \neq j$, so the pairwise angles are $\arccos(-1/3) \approx 109.5^\circ$. The great circles $\mathbf{n}_i \cdot \mathbf{d} = 0$ intersect pairwise; the three intersection points lying in $\mathbf{n}_4 \cdot \mathbf{d} > 0$ are

$$\mathbf{a} = \frac{1}{\sqrt{2}}(0, 1, -1), \quad \mathbf{b} = \frac{1}{\sqrt{2}}(1, 0, -1), \quad \mathbf{c} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

All pairwise dot products equal $1/2$, so all side-lengths of the spherical triangle E_4 are $\pi/3$. By the spherical law of cosines the interior angles are each $\arccos(1/3)$, and the spherical excess (= solid angle) is

$$|E_k| = 3 \arccos\left(\frac{1}{3}\right) - \pi \approx 0.551 \text{ sr}.$$

In particular, $|E_k|/(4\pi) \approx 0.044$, far smaller than $1/4$. (The Voronoi cells of $\mathbf{n}_i$ on S^2 , which equal the stability regions S_i at the centroid, each have area π by symmetry; the exclusive cones E_k are much smaller subsets of these cells.)

6 Two-Material Density Distributions

6.1 The density field

We consider density distributions of the following form. The interior of P is partitioned into two measurable regions Σ_H and $\Sigma_L = P \setminus \Sigma_H$, with densities ρ_H and ρ_L respectively, where $\rho_H > \rho_L > 0$. The center of mass is:

$$\mathbf{c}(\Sigma_H) = \frac{\rho_H \int_{\Sigma_H} \mathbf{x} dV + \rho_L \int_{\Sigma_L} \mathbf{x} dV}{\rho_H |\Sigma_H| + \rho_L |\Sigma_L|}, \quad (7)$$

where $|\cdot|$ denotes volume. Define the density ratio $r = \rho_H/\rho_L > 1$ and the volume of the polytope $V_P = |P|$. Setting $\alpha = |\Sigma_H|/V_P \in (0, 1)$ and letting $\bar{\mathbf{x}}_H$ and $\bar{\mathbf{x}}_L$ denote the centroids of Σ_H and Σ_L respectively:

$$\mathbf{c} = \frac{r\alpha \bar{\mathbf{x}}_H + (1 - \alpha) \bar{\mathbf{x}}_L}{r\alpha + (1 - \alpha)}. \quad (8)$$

6.2 Achievability of arbitrary centers of mass

Theorem 17 (Approximate reachability). *For any convex polytope P and any point $\mathbf{c}^* \in \text{int}(P)$, and any $\delta > 0$, there exists a threshold ratio r^* such that for all $r > r^*$, there exists a two-material partition (Σ_H, Σ_L) with density ratio r whose center of mass $\mathbf{c}$ satisfies $\|\mathbf{c} - \mathbf{c}^*\| < \delta$.*

Proof. Since $\mathbf{c}^* \in \text{int}(P)$, there exists $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{c}^*) \subset P$. Set $\Sigma_H = B_\varepsilon(\mathbf{c}^*)$, so $\bar{\mathbf{x}}_H = \mathbf{c}^*$ (centroid of a ball centered at $\mathbf{c}^*$), and let $\bar{\mathbf{x}}_L$ be the centroid of $\Sigma_L = P \setminus \overline{\Sigma_H}$. Let $\alpha = |B_\varepsilon(\mathbf{c}^*)|/V_P > 0$. From equation (8):

$$\mathbf{c}(r) - \mathbf{c}^* = \frac{(1 - \alpha)(\bar{\mathbf{x}}_L - \mathbf{c}^*)}{r\alpha + (1 - \alpha)}.$$

As $r \rightarrow \infty$:

$$\|\mathbf{c}(r) - \mathbf{c}^*\| = \frac{(1 - \alpha)\|\bar{\mathbf{x}}_L - \mathbf{c}^*\|}{r\alpha + (1 - \alpha)} \rightarrow 0.$$

Thus for any $\delta > 0$, there exists r^* such that $r > r^*$ implies $\|\mathbf{c}(r) - \mathbf{c}^*\| < \delta$. $\square$

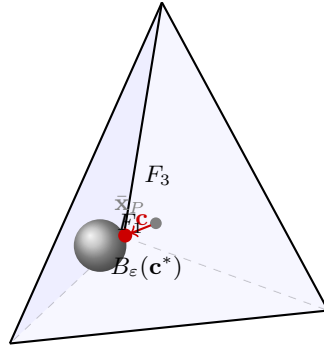


Figure 10: A non-degenerate tetrahedron with a tungsten ball $B_\varepsilon(\mathbf{c}^*)$ (gray sphere) embedded in the interior. The center of mass shifts from the geometric centroid $\bar{\mathbf{x}}_P$ toward the ball, reaching a position $\mathbf{c}$ that can be made arbitrarily close to any target $\mathbf{c}^*$ by increasing the density ratio r (Theorem 17).

Remark 18. Theorem 17 shows that the density ratio r controls the “reachable set” of centers of mass. For $r = 1$ (uniform density), only the centroid $\bar{\mathbf{x}}_P$ is achievable. As $r \rightarrow \infty$, the entire interior becomes reachable. In practice, a resin/tungsten combination gives $r \approx 16$, which suffices for substantial displacement of the center of mass.

6.3 The embedded sphere model

The general two-material partition (Σ_H, Σ_L) has infinite-dimensional freedom in the choice of Σ_H . A natural constructive special case is to take Σ_H to be a single ball: embed a sphere of dense material (e.g. tungsten, density ρ_H) inside a light body (e.g. resin, density ρ_L). This model has finitely many parameters and admits a clean closed-form analysis.

Let $B_a(\mathbf{s})$ denote the closed ball of radius $a > 0$ centered at $\mathbf{s} \in \text{int}(P)$, subject to the constraint $B_a(\mathbf{s}) \subset P$ (the ball fits entirely inside the polytope). The feasible set of centers is the *inner parallel body*:

$$P_{-a} = P \ominus B_a = \{\mathbf{x} \in P : d(\mathbf{x}, \partial P) \geq a\}, \quad (9)$$

where $\ominus$ denotes the Minkowski difference. The maximum feasible radius is the inradius $a_{\max} = \max_{\mathbf{x} \in P} d(\mathbf{x}, \partial P)$.

The composite body has density ρ_H on $B_a(\mathbf{s})$ and ρ_L on $P \setminus B_a(\mathbf{s})$. By superposition, this equals the uniform body P at density ρ_L plus an excess density $\Delta\rho = \rho_H - \rho_L$ on the ball. Since the ball is symmetric, its centroid is $\mathbf{s}$. Writing $V_P = \text{Vol}(P)$ and $V_a = \frac{4}{3}\pi a^3$, the component masses are:

$$\begin{aligned} M_P &= \rho_L V_P, \\ M_{\text{ball}} &= (\rho_H - \rho_L) V_a, \end{aligned}$$

and the composite center of mass is:

$$\mathbf{c}(\mathbf{s}, a) = \frac{M_P \bar{\mathbf{x}}_P + M_{\text{ball}} \mathbf{s}}{M_P + M_{\text{ball}}} = (1 - \lambda) \bar{\mathbf{x}}_P + \lambda \mathbf{s}, \quad (10)$$

where $\bar{\mathbf{x}}_P$ is the centroid of P and

$$\lambda = \lambda(a) = \frac{(r - 1) V_a}{V_P + (r - 1) V_a} \in (0, 1), \quad (11)$$

with $r = \rho_H / \rho_L > 1$ the density ratio as before.

The center of mass (10) is a convex combination of the polytope centroid $\bar{\mathbf{x}}_P$ and the sphere center $\mathbf{s}$, with weight λ depending on a and r but not on $\mathbf{s}$.

Proposition 19 (Affine structure). *For fixed sphere radius a (and hence fixed λ), the map $\mathbf{s} \mapsto \mathbf{c}(\mathbf{s}, a)$ is an affine contraction—a homothety centered at $\bar{\mathbf{x}}_P$ with ratio λ :*

$$\mathbf{c} - \bar{\mathbf{x}}_P = \lambda (\mathbf{s} - \bar{\mathbf{x}}_P). \quad (12)$$

As $\mathbf{s}$ ranges over the inner parallel body P_{-a} , the reachable set of centers of mass is a scaled copy of P_{-a} shrunk by factor λ toward $\bar{\mathbf{x}}_P$:

$$\mathcal{R}_a = (1 - \lambda) \bar{\mathbf{x}}_P + \lambda P_{-a}.$$

Proof. Immediate from (10). □

The model has 4 parameters: $\mathbf{s} \in \mathbb{R}^3$ (3 DOF) and $a \in \mathbb{R}_{>0}$ (1 DOF). However, the image $\mathbf{c}(\mathbf{s}, a)$ lies in $\mathbb{R}^3$, and there is a redundancy: along any ray from $\bar{\mathbf{x}}_P$, both $|\mathbf{s} - \bar{\mathbf{x}}_P|$ and a (through λ) control $|\mathbf{c} - \bar{\mathbf{x}}_P|$. The effective dimension of the reachable set is 3.

Varying both $\mathbf{s}$ and a , the full reachable set is

$$\mathcal{R} = \bigcup_{0 < a < a_{\max}} \mathcal{R}_a = \bigcup_{0 < a < a_{\max}} \{(1 - \lambda(a)) \bar{\mathbf{x}}_P + \lambda(a) \mathbf{s} : \mathbf{s} \in P_{-a}\}. \quad (13)$$

Remark 20 (Inverse formula for sphere placement). Given a target center of mass $\mathbf{c}^*$ and a choice of radius a (which determines λ), the required sphere center is:

$$\mathbf{s}^* = \bar{\mathbf{x}}_P + \frac{1}{\lambda} (\mathbf{c}^* - \bar{\mathbf{x}}_P). \quad (14)$$

This is feasible ($\mathbf{s}^* \in P_{-a}$) provided $\mathbf{c}^*$ is not too far from $\bar{\mathbf{x}}_P$ relative to λ and the inradius of P . For λ close to 1 (large density ratio or large sphere), $\mathbf{s}^* \approx \mathbf{c}^*$; for small λ , the required $\mathbf{s}^*$ may fall outside P_{-a} .

Remark 21 (Practical advantages). The embedded sphere model is particularly amenable to manufacturing: drill or mold a spherical cavity in the resin body, then insert a tungsten ball bearing. The design problem reduces to choosing $(a, \mathbf{s})$ —a sphere radius and a position—rather than an arbitrary volumetric partition. The affine structure (12) makes the relationship between sphere placement and center-of-mass displacement transparent and invertible.

7 The Fairness Locus

Definition 22. Let P be a convex polytope with N faces. The *fairness locus* of P is:

$$\mathcal{F}(P) = \{\mathbf{c} \in \text{int}(P) : p_1(\mathbf{c}) = p_2(\mathbf{c}) = \cdots = p_N(\mathbf{c}) = \frac{1}{N}\}.$$

Since $\sum p_i = 1$, the condition $p_i(\mathbf{c}) = 1/N$ for all i constitutes $N - 1$ independent equations in the three coordinates of $\mathbf{c}$.

Proposition 23 (Dimensional analysis). *For a convex polytope P with $N \geq 4$ faces, assume that $\mathbf{c}$ ranges over an open set of points in general position (Definition 10). Assume further that the map $\mathbf{c} \mapsto \mathbf{p}(\mathbf{c}) = (p_1(\mathbf{c}), \dots, p_N(\mathbf{c}))$ is smooth in this domain and that the target point $(1/N, \dots, 1/N)$ is a regular value of the restriction of $\mathbf{p}$ to points in general position. Then:*

- (i) *When $N = 4$: the fairness locus $\mathcal{F}(P)$ is a smooth manifold of dimension $3 - (N - 1) = 4 - N = 0$, i.e., a discrete set of points.*
- (ii) *When $N > 4$: the fairness locus is generically empty (dimension $4 - N < 0$), unless P has exceptional symmetry.*

Proof. Define $\Phi : U \rightarrow \mathbb{R}^{N-1}$ where $U \subseteq \text{int}(P)$ is the open set of points in general position, and

$$\Phi(\mathbf{c}) = (p_1(\mathbf{c}) - 1/N, \dots, p_{N-1}(\mathbf{c}) - 1/N).$$

The fairness locus in U is $\Phi^{-1}(\mathbf{0})$.

The domain U is 3-dimensional. By assumption, $(1/N, \dots, 1/N)$ is a regular value of Φ , meaning that for every $\mathbf{c} \in \Phi^{-1}(\mathbf{0}) \cap U$, the Jacobian $D\Phi(\mathbf{c})$ has rank $N - 1$.

By the preimage theorem (also called the regular value theorem), if $\mathbf{0}$ is a regular value of Φ , then $\Phi^{-1}(\mathbf{0})$ is a smooth submanifold of U of dimension $\dim(U) - \text{rank}(D\Phi) = 3 - (N - 1) = 4 - N$.

For $N = 4$, we have $4 - N = 0$, so the fairness locus is a discrete set of points.

For $N > 4$, we have $4 - N < 0$, which is impossible for a submanifold. By the transversality theorem, if the map $\mathbf{p}$ is transverse to the target point (or equivalently, the target is a regular value), then the preimage of a codimension- $(N - 1)$ point in a 3-dimensional domain is generically empty when $N - 1 > 3$. $\square$

Remark 24. The case $N = 4$ is particularly clean: the fairness locus is generically a discrete (finite) set of points. For a tetrahedron, we expect finitely many center-of-mass positions achieving exact fairness.

Remark 25. The word “generic” denotes the typical case after any exceptional symmetries are broken. A *generic* polytope is one with no special symmetries: its symmetry group is trivial. Highly symmetric polytopes may have fairness loci of unexpectedly large dimension. A regular cube with $N = 6$ has $\mathcal{F}(P) \ni \bar{\mathbf{x}}_P$ (the centroid) by symmetry, despite the dimensional analysis predicting dimension $4 - 6 = -2$. Proposition 24 describes the typical case after symmetry is generically broken.

8 Existence Results

8.1 Tetrahedra

Definition 26 (Non-degenerate tetrahedron). A tetrahedron T is *non-degenerate* if its four vertices are not coplanar (i.e., T has positive volume), and no three vertices are collinear. Equivalently, T is a full-dimensional simplex in $\mathbb{R}^3$.

Theorem 27 (Fair tetrahedra). *For any non-degenerate tetrahedron $T \subset \mathbb{R}^3$ (Definition 27), the fairness locus $\mathcal{F}(T)$ is non-empty. That is, there exists $\mathbf{c}^* \in \text{int}(T)$ such that all four faces have equal rest probability $1/4$.*

Proof. Let T have faces $F_1, \dots, F_4$ with opposite vertices $v_1, \dots, v_4$ (vertex v_i is opposite face F_i). Define:

$$f : \text{int}(T) \rightarrow \mathbb{R}^3, \quad f(\mathbf{c}) = (p_1(\mathbf{c}) - \tfrac{1}{4}, p_2(\mathbf{c}) - \tfrac{1}{4}, p_3(\mathbf{c}) - \tfrac{1}{4}).$$

We must show $f^{-1}(\mathbf{0}) \neq \emptyset$. We use a topological degree argument.

Step 1: Boundary nonvanishing. By Lemma 15, f is nonvanishing near face interiors and edge interiors of ∂T for any non-degenerate tetrahedron. For vertices, the lemma gives nonvanishing under the condition $|E_k| \neq \pi$, which we verify for T_0 in Step 2 (Remark 16).

Step 2: Regular tetrahedron and Jacobian nonsingularity. Consider a regular tetrahedron T_0 with vertices at:

$$v_1 = (1, 1, 1), \quad v_2 = (1, -1, -1), \quad v_3 = (-1, 1, -1), \quad v_4 = (-1, -1, 1).$$

These vertices form a regular tetrahedron: all edges have the same length $\sqrt{8}$, and the tetrahedron is invariant under the tetrahedral symmetry group T_d of order 24. The centroid is:

$$\bar{\mathbf{x}} = \frac{1}{4}(v_1 + v_2 + v_3 + v_4) = \frac{1}{4}(0, 0, 0) = \mathbf{0}.$$

By symmetry, the four faces are equivalent under T_d , so the stability regions at $\mathbf{0}$ are congruent spherical triangles of equal area:

$$p_1(\mathbf{0}) = p_2(\mathbf{0}) = p_3(\mathbf{0}) = p_4(\mathbf{0}) = \tfrac{1}{4}.$$

Thus $f_0(\mathbf{0}) = \mathbf{0}$.

The *inward* face normals of the regular tetrahedron (pointing from each face toward the interior) are:

$$\hat{\mathbf{n}}_1 = (1, 1, 1)/\sqrt{3}, \quad \hat{\mathbf{n}}_2 = (1, -1, -1)/\sqrt{3}, \quad \hat{\mathbf{n}}_3 = (-1, 1, -1)/\sqrt{3}, \quad \hat{\mathbf{n}}_4 = (-1, -1, 1)/\sqrt{3}.$$

(Verification: face F_1 has vertices v_2, v_3, v_4 ; the cross product $(v_3 - v_2) \times (v_4 - v_2) \propto (1, 1, 1)$ points from F_1 toward the opposite vertex $v_1 = (1, 1, 1)$, i.e., into the tetrahedron.) The outward normals are $\mathbf{n}_i = -\hat{\mathbf{n}}_i$.

At the centroid $\mathbf{0}$, by symmetry of the stability regions, the gradient of each probability function p_i points toward the supporting plane of F_i , i.e., in the outward normal direction $\mathbf{n}_i = -\hat{\mathbf{n}}_i$. Moving $\mathbf{c}$ in direction $-\hat{\mathbf{n}}_i$ (toward face F_i , away from the opposite vertex v_i) decreases $|w_i|$, bringing $\mathbf{c}$ closer to F_i , which causes S_i to grow (by Lemma 14):

$$\nabla p_i(\mathbf{0}) \propto -\hat{\mathbf{n}}_i.$$

The 3×3 Jacobian matrix $Df_0(\mathbf{0}) = D(p_1, p_2, p_3)(\mathbf{0})$ has rows proportional to $-\hat{\mathbf{n}}_i$. Since $\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2, \hat{\mathbf{n}}_3$ are linearly independent, so are $-\hat{\mathbf{n}}_1, -\hat{\mathbf{n}}_2, -\hat{\mathbf{n}}_3$:

$$\det[-\hat{\mathbf{n}}_1, -\hat{\mathbf{n}}_2, -\hat{\mathbf{n}}_3] = (-1)^3 \det[\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2, \hat{\mathbf{n}}_3] = - \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \end{vmatrix} = -4 \neq 0.$$

Thus $Df_0(\mathbf{0})$ has rank 3 and is nonsingular.

Step 3: Homotopy and degree preservation. Let T_0 be the regular tetrahedron with vertices $v_i^{(0)}$ as above, and $T_1 = T$ the given non-degenerate tetrahedron with vertices $v_i^{(1)}$. If T_1 has the opposite orientation from T_0 (i.e., $\det[v_2^{(1)} - v_1^{(1)}, v_3^{(1)} - v_1^{(1)}, v_4^{(1)} - v_1^{(1)}]$ has the opposite sign), relabel two vertices of T_1 to make it positively oriented; the fairness locus is invariant under vertex relabeling.

The space of non-degenerate positively-oriented tetrahedra, parameterized by vertex positions in $(\mathbb{R}^3)^4$ subject to $\det[v_2 - v_1, v_3 - v_1, v_4 - v_1] > 0$, is path-connected. (It is homeomorphic to $GL^+(3, \mathbb{R}) \times \mathbb{R}^3$ via the affine map determined by the vertices, and $GL^+(3, \mathbb{R})$ is connected.)

Choose any continuous path $\{T_t\}_{t \in [0,1]}$ of non-degenerate positively-oriented tetrahedra with T_0 the regular tetrahedron and $T_1 = T$. Define the affine pullback $\phi_t : T_0 \rightarrow T_t$ by $\phi_t(\mathbf{x}) = A(t)\mathbf{x} + \mathbf{b}(t)$, where $A(t)$ and $\mathbf{b}(t)$ are determined by $\phi_t(v_i^{(0)}) = v_i(t)$ for $i = 1, \dots, 4$. Since each T_t is non-degenerate, $A(t)$ is invertible and ϕ_t is a diffeomorphism $T_0 \rightarrow T_t$ depending continuously on t , with $\phi_0 = \text{id}$.

We show that the set

$$S = \{t \in [0, 1] : \mathcal{F}(T_t) \neq \emptyset\}$$

is non-empty and *relatively closed* in $[0, 1]$, and that S contains a neighborhood of 0. Since $[0, 1]$ is connected, $S = [0, 1]$ once we show S is also open, but the closedness argument alone propagates existence to $t = 1$ as follows.

$0 \in S$: The centroid $\mathbf{0}$ is a fair center of T_0 (by symmetry, $p_i(\mathbf{0}) = 1/4$ for all i).

S is open: At $t_0 \in S$, let $\mathbf{c}_0 \in \text{int}(T_{t_0})$ be a fair center. If $Df_{t_0}(\mathbf{c}_0)$ is nonsingular, the implicit function theorem gives a zero of f_t near $\mathbf{c}_0$ for t near t_0 . In the degenerate case, a topological degree argument extends the result: there exists an open set $U \ni \mathbf{c}_0$ in $\text{int}(T_{t_0})$ with $f_{t_0} \neq \mathbf{0}$ on ∂U and $\deg(f_{t_0}, U, \mathbf{0}) \neq 0$, so by homotopy invariance of the degree the zero persists for t near t_0 .⁵

S is closed: Let $t_n \in S$ with $t_n \rightarrow t^*$, and let $\mathbf{c}_n \in \text{int}(T_{t_n})$ satisfy $f_{t_n}(\mathbf{c}_n) = \mathbf{0}$. The sequence $\mathbf{x}_n = \phi_{t_n}^{-1}(\mathbf{c}_n) \in \text{int}(T_0)$ lies in the compact set T_0 , so a subsequence converges: $\mathbf{x}_{n_j} \rightarrow \mathbf{x}^* \in T_0$. We claim $\mathbf{x}^* \in \text{int}(T_0)$. If not, $\mathbf{x}^*$ lies on ∂T_0 and $\mathbf{c}_{n_j} = \phi_{t_{n_j}}(\mathbf{x}_{n_j}) \rightarrow \phi_{t^*}(\mathbf{x}^*) \in \partial T_{t^*}$. But Lemma 15 (face and edge cases⁶) gives $\liminf \|f_{t^*}(\mathbf{c})\| > 0$ as $\mathbf{c} \rightarrow \phi_{t^*}(\mathbf{x}^*)$ from $\text{int}(T_{t^*})$. Since the probability functions depend continuously on the tetrahedron's vertices (the stability regions vary continuously with the face normals and $\mathbf{c}$), $f_{t_{n_j}}(\mathbf{c}_{n_j}) \rightarrow 0$ implies $f_{t^*}(\phi_{t^*}(\mathbf{x}^*)) = 0$, contradicting the nonvanishing. Therefore $\mathbf{x}^* \in \text{int}(T_0)$ and $\mathbf{c}^* = \phi_{t^*}(\mathbf{x}^*) \in \text{int}(T_{t^*})$ with $f_{t^*}(\mathbf{c}^*) = 0$, so $t^* \in S$.

Since S is non-empty, open, and closed in $[0, 1]$, we have $S = [0, 1]$.

Step 4: Conclusion. Since $1 \in S$, the tetrahedron $T_1 = T$ has a fair center $\mathbf{c}^* \in \text{int}(T)$ with $p_i(\mathbf{c}^*) = 1/4$ for all i . Thus $\mathcal{F}(T) \neq \emptyset$. $\square$

Corollary 28. *For any non-degenerate tetrahedron and any $\varepsilon > 0$, there exists a two-material density distribution such that $|p_i - 1/4| < \varepsilon$ for all i .*

Proof. Theorem 28 provides a fair center-of-mass position $\mathbf{c}^* \in \text{int}(T)$. By continuity of $p_i(\mathbf{c})$, there exists $\delta > 0$ such that $\|\mathbf{c} - \mathbf{c}^*\| < \delta$ implies $|p_i(\mathbf{c}) - 1/4| < \varepsilon$ for all i . By Theorem 17, for sufficiently large r , a two-material partition achieves $\|\mathbf{c} - \mathbf{c}^*\| < \delta$. $\square$

⁵The nonvanishing on ∂U uses the fact that zeros of f_{t_0} are isolated, which holds generically by Sard's theorem. At the non-generic t -values where zeros coalesce, a more delicate argument involving the parametric Sard theorem is needed; we omit the details.

⁶The vertex case requires $|E_k(T_{t^*})| \neq \pi$; if this fails, a separate perturbation argument is needed. For the regular tetrahedron and nearby tetrahedra, the condition holds (Remark 16).

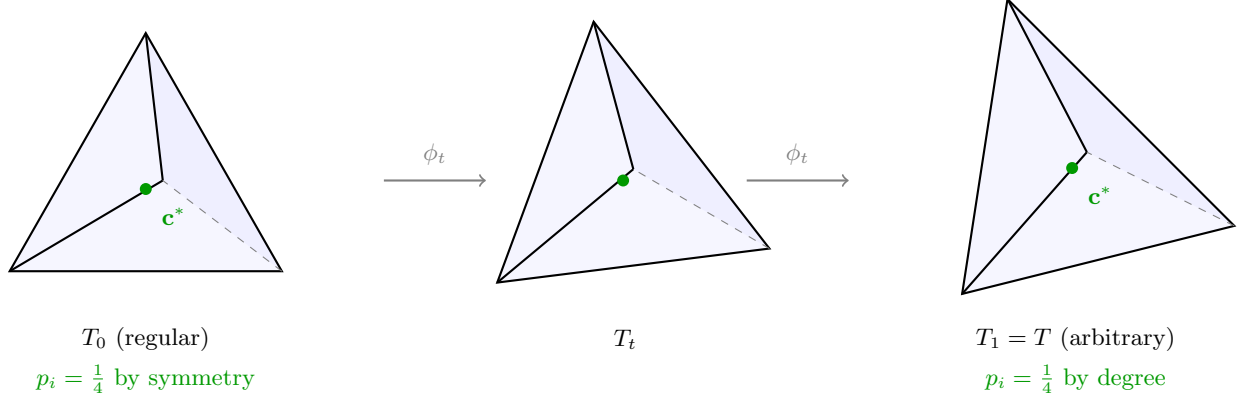


Figure 11: The homotopy argument of Theorem 28. A continuous path of non-degenerate tetrahedra $\{T_t\}_{t \in [0,1]}$ connects the regular tetrahedron T_0 (where fairness holds by symmetry at the centroid) to an arbitrary tetrahedron $T_1 = T$. The affine pullback $\phi_t : T_0 \rightarrow T_t$ transfers the fairness map to a fixed domain; since the Brouwer degree is nonzero at $t = 0$ and never vanishes on the boundary, it persists to $t = 1$, guaranteeing a fair center of mass $\mathbf{c}^*$ for every non-degenerate tetrahedron.

8.2 Beyond four faces

For polytopes with $N \geq 5$ faces, Proposition 24 shows that exact fairness via center-of-mass manipulation alone is generically impossible: the system of $N - 1$ equations in 3 unknowns is overdetermined.

However, several avenues extend the achievable regime.

Face grouping. If the polytope has N faces but we seek a fair die with $K < N$ outcomes, we may assign the same label to multiple faces. The fairness condition becomes $K - 1$ equations (the grouped probabilities must each equal $1/K$). For $K \leq 4$, this is generically solvable by dimensional analysis (3 unknowns, $K - 1 \leq 3$ equations).

Conjecture 29. For any convex polytope with $N \geq 4$ faces and any $K \leq 4$, there generically exists a grouping of faces into K classes and a center-of-mass position such that each class has rest probability $1/K$. Here, “generically” means for a convex polytope with no exceptional symmetries (i.e., a generic polytope in the sense of Proposition 24).

Remark 30. The dimensional count from Proposition 24 suggests that Conjecture 30 should hold: $K - 1 \leq 3$ equations in 3 unknowns is not overdetermined, so solutions generically exist. We have proved this rigorously for $K = 4$ and $N = 4$ (Theorem 28) using a topological degree argument. Extending this argument to general N with face grouping would require establishing the appropriate boundary non-vanishing condition (analogous to Lemma 15) for the grouped probability map, which we leave as an open problem.

Approximate fairness. For $N > 4$, we can minimize the deviation from fairness. Define:

$$\delta(P) = \min_{\mathbf{c} \in \text{int}(P)} \max_{1 \leq i \leq N} \left| p_i(\mathbf{c}) - \frac{1}{N} \right|. \quad (15)$$

This is the *minimum unfairness* of P , a geometric invariant. For symmetric polytopes, $\delta(P) = 0$. For mildly asymmetric polytopes (small perturbations of symmetric ones), $\delta(P)$ is small by continuity.

Inertia tensor extension. The stability model of Section 5.2 assumes the final orientation is uniformly random, making the rest probabilities depend only on $\mathbf{c}$. In a more realistic physical model, the die’s tumbling dynamics depend on the full *inertia tensor* $\mathbf{I} \in \text{Sym}_3^+$, the space of 3×3 symmetric positive-definite matrices, which has 6 independent components.

A two-material density distribution determines both $\mathbf{c}$ (3 parameters) and $\mathbf{I}$ (6 parameters), giving 9 degrees of design freedom. Under a physics model in which rest probabilities depend on both $\mathbf{c}$ and $\mathbf{I}$, the fairness condition ($N - 1$ equations in 9 unknowns) is generically solvable for $N \leq 10$. The following remark is speculative and describes a promising direction for future work.

Remark 31 (Inertia tensor design). Under a physically realistic rest probability model $p_i(\mathbf{c}, \mathbf{I})$ that accounts for both the center of mass and inertia tensor, for any convex polytope with $N \leq 10$ stable resting faces, one might expect that a two-material density distribution generically achieves exact fairness. The heuristic justification is dimensional: the space of achievable $(\mathbf{c}, \mathbf{I})$ pairs (9-dimensional) exceeds the dimension of the fairness constraint (8 independent equations for $N = 10$). However, this is speculative and requires validation with a detailed physical model of die tumbling dynamics.

Since most natural objects have fewer than 10 distinct stable resting configurations, this suggests that density optimization can achieve fairness for a very wide class of shapes in practice.

9 Optimization Framework

For polytopes with $N > 4$ faces (where exact fairness via the center-of-mass model is not guaranteed), or for polytopes where the fairness locus is difficult to compute analytically, we formulate the density design as a numerical optimization problem.

9.1 Discretization

Partition the interior of P into K voxels with volumes $v_1, \dots, v_K$ and centroids $\mathbf{x}_1, \dots, \mathbf{x}_K$. Each voxel is assigned density $\rho_k \in \{\rho_L, \rho_H\}$ (binary) or, in a relaxed formulation, $\rho_k \in [\rho_L, \rho_H]$ (continuous). The center of mass is:

$$\mathbf{c}(\boldsymbol{\rho}) = \frac{\sum_{k=1}^K \rho_k v_k \mathbf{x}_k}{\sum_{k=1}^K \rho_k v_k}.$$

9.2 Objective

Given a target probability distribution $\mathbf{q} = (q_1, \dots, q_N)$ on the N faces (with $q_i = 1/N$ for fair dice, and $\sum_{i=1}^N q_i = 1$), minimize:

$$J(\boldsymbol{\rho}) = \sum_{i=1}^N (p_i(\mathbf{c}(\boldsymbol{\rho})) - q_i)^2. \quad (16)$$

Under the stability model, $p_i(\mathbf{c})$ is the normalized solid angle of the stability region of face i , which depends smoothly on $\mathbf{c}$ (away from degenerate configurations where $\mathbf{c}$ lies on a stability boundary). Since $\mathbf{c}$ depends smoothly on $\boldsymbol{\rho}$, the objective J is smooth and amenable to gradient-based optimization.

9.3 Gradient computation

By the chain rule:

$$\frac{\partial J}{\partial \rho_k} = \sum_{i=1}^N 2(p_i - q_i) \sum_{j=1}^3 \frac{\partial p_i}{\partial c_j} \frac{\partial c_j}{\partial \rho_k}.$$

The partial derivatives $\partial c_j / \partial \rho_k$ follow from (7) by direct differentiation. Let $M = \sum_{\ell} \rho_{\ell} v_{\ell}$ (total mass). By the quotient rule:

$$\frac{\partial c_j}{\partial \rho_k} = \frac{v_k x_{k,j} \cdot M - (\sum_{\ell} \rho_{\ell} v_{\ell} x_{\ell,j}) \cdot v_k}{M^2} = \frac{v_k}{M} (x_{k,j} - c_j),$$

where $x_{k,j}$ is the j -th coordinate of the k -th voxel centroid. In summary:

$$\frac{\partial c_j}{\partial \rho_k} = \frac{v_k}{M} (x_{k,j} - c_j). \quad (17)$$

For the partial derivatives $\partial p_i / \partial c_j$: geometrically, perturbing $\mathbf{c}$ shifts the power weights (5) and the boundaries of the stability regions. As $\mathbf{c}$ moves in direction $\mathbf{u}$, the boundary arcs ∂S_i (which separate S_i from adjacent stability regions) shift, changing the area of S_i .

Remark 32. A natural candidate formula for $\partial p_i / \partial c_j$ is:

$$\frac{\partial p_i}{\partial c_j} = \frac{1}{4\pi} \oint_{\partial S_i} n_j d\sigma, \quad (18)$$

where the integral is over the boundary arcs of S_i and $d\sigma$ is arc length on S^2 . However, the correct Leibniz-rule derivation requires tracking the velocity of the boundary ∂S_i under perturbation of $\mathbf{c}$, not merely the boundary normal. The formula (18) is therefore best treated as a heuristic; in practice, it is more reliable to compute $\partial p_i / \partial c_j$ numerically via finite differences or automatic differentiation.

For automatic differentiation, the computation of $p_i(\mathbf{c})$ can be implemented in a differentiable framework (e.g., JAX), enabling end-to-end gradient descent on J .

9.4 Binary constraint

In the two-material model, each $\rho_k \in \{\rho_L, \rho_H\}$. This is a combinatorial constraint. Two practical approaches:

Relaxation and rounding. Optimize over $\rho_k \in [\rho_L, \rho_H]$ continuously, then round each ρ_k to the nearest endpoint. Adjust a small number of voxels to fine-tune the center of mass.

Evolutionary search. Represent the partition as a binary string $\sigma \in \{0, 1\}^K$ (where $\sigma_k = 1$ means voxel k is heavy) and use CMA-ES or a genetic algorithm to minimize J .

9.5 Designer distributions

The framework is not limited to fair dice. By choosing $\mathbf{q} \neq (1/N, \dots, 1/N)$, one can design dice that produce *any* desired probability distribution. For example:

- (a) A regular dodecahedron ($N = 12$) with faces labeled 2 through 12, density-optimized to produce the distribution of the sum of two standard six-sided dice. Since the dodecahedron has 12 faces but only 11 distinct labels $(2, 3, \dots, 12)$, one label must appear on two faces. Following the face-grouping framework of Conjecture 30, face 7 is assigned to two faces (each contributing probability $3/36$, for a combined $6/36$); faces 2 and 12 have probability $1/36$ each.

- (b) A standard cube density-optimized to produce a Zipf distribution, where face k has probability proportional to $1/k$.

10 Discussion

10.1 Physical realizability

The two-material model is directly realizable via modern manufacturing. A resin body ($\rho_L \approx 1.2$ g/cm³) with tungsten inserts ($\rho_H \approx 19.3$ g/cm³) gives a density ratio of approximately 16. By Theorem 17, this suffices to place the center of mass over a wide range of interior positions.

Multi-material 3D printing (e.g., stereolithography with post-inserted metal components) enables fabrication. The voxel partition from Section 9 translates directly into a CAD model with cavities for tungsten inserts.

10.2 The dimensional barrier and its circumvention

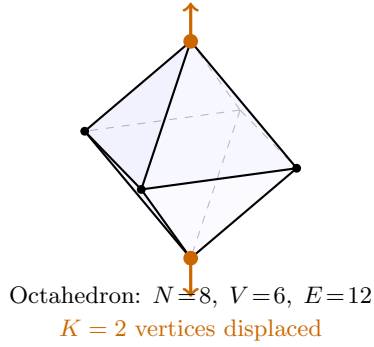
Proposition 24 identifies a fundamental dimensional barrier: center-of-mass manipulation provides 3 degrees of freedom, which is insufficient for exact fairness when $N \geq 5$. Four approaches circumvent this:

- (i) *Face grouping* (Conjecture 30): reduce the number of distinct outcomes to $K \leq 4$.
- (ii) *Vertex displacement* (Section 2, Theorem 3): for simplicial polytopes, displace vertices (“toothpicks at the corners”) and embed a dense sphere. The parameter surplus is $N/2 + 4$, growing linearly with N . This operates entirely within the standard stability model—no modification of the physics, only of the shape.
- (iii) *Inertia tensor exploitation* (Remark 32): use the full 9 degrees of freedom available in the density distribution, under a physics model sensitive to the inertia tensor (speculative).
- (iv) *General shape co-optimization*: if the boundary shape is an unconstrained design variable, the problem gains effectively infinite-dimensional freedom and the dimensional barrier vanishes.

Approach (ii) is the most concrete: a physically realizable construction (insert rods at corners, embed a ball bearing) that operates within the standard stability model. The parameter surplus (Theorem 3) is a necessary condition for solvability; the numerical $d5$ example (Section 4) provides evidence that the surplus translates to actual solutions, but a rigorous existence proof for general $N > 4$ remains open. Approach (iv) subsumes (ii) but is less constructive.

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Parameter count

Vertex positions:	$3V$
Sphere center:	3
Rigid motions:	-6

Effective parameters: $3V - 3$

Fairness constraints: $N - 1$

Surplus = $3V - N - 2 = \frac{N}{2} + 4$

Octahedron: $18 + 3 - 6 = 15$ params, 7 constr.
 Surplus = 8. Only $K \geq 2$ vertices needed.

Figure 12: Parameter count for vertex displacement fairness (Theorem 3). Left: an octahedron ($N = 8$ faces, $V = 6$ vertices, all faces triangular) with two vertices displaced by toothpicks (orange arrows). Right: $3V + 3$ raw parameters minus 6 for rigid motions gives $3V - 3$ effective degrees of freedom, exceeding the $N - 1$ fairness constraints by $N/2 + 4$ (Euler's formula: $V = 2 + N/2$).

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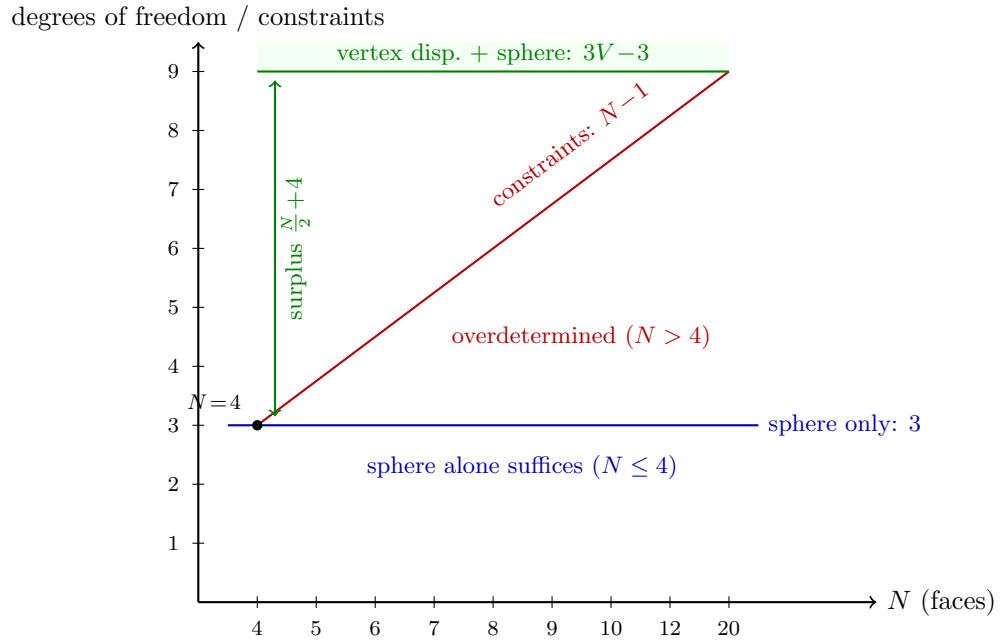


Figure 13: The dimensional barrier and its resolution. The sphere-only model (blue, 3 DOF) meets the $N - 1$ constraints (red) at $N = 4$; for $N > 4$ the system is overdetermined. Vertex displacement combined with sphere embedding (green region) provides $3V - 3$ effective parameters, which by Euler's formula always exceeds $N - 1$ by at least $N/2 + 4 \geq 6$, resolving the barrier for all N .